Trainings ▾

General Information

Viscosity is a measure of the resistance to flow oil. The viscosity characteristics of all fluids are affected by temperature. Multigrade oil (e.g. 15W-40) viscosities tend to be less sensitive to temperature changes than monograde oil (e.g. Society of Automotive Engineers (SAE) 40) because of the addition of viscosity improvers in their formulation. A correctly formulated multigrade oil **not only** protects the engine at startup, but also throughout the life of the engine.

The selection of oil of the correct viscosity is extremely important for optimum performance and for maximum engine life. If the oil is too viscous, engine drag is increased with the following effects:

- Engine is difficult to start.
- Engine power output is reduced.
- Engine cooling is reduced.
- Internal wear is increased.
- Engine parts run hotter.
- Fuel consumption is increased.

If the oil viscosity is too low, the engine experiences:

- Increased wear from metal-to-metal contact.
- Increased oil consumption and leakage.
- Increased engine noise.

A specific viscosity advertised (e.g. SAE 15W-40) on a product is a viscosity designation **only**. This designation alone does **not** imply that the product meets the Cummins Inc. requirements above.

Some oil suppliers might claim significant fuel economy improvements with lower viscosity oils. Make sure that the oil's viscosity grade is permitted in the owners manual and the oil is registered to a Cummins Engineering Standard recommended for your engine. Otherwise, obtain the oil supplier's commitment that they will assume warranty liability, or do **not** use the oil.

Note : For K50, K38, QSK38, QSK50, QSK45, QSK60, and QSK78, consider use of 15W-40 viscosity grade for ambient temperature down to -20ºC (-4ºF) for ambient temperature below -20ºC (-4ºF) use either 0W-40, 5W-40, or 10W-40 depending on the minimum ambient temperature, see table below.

Viscosity Recommendations

General recommendations for oil viscosity grade versus low ambient temperatures can be found in Figure 1.

Not all viscosity grades are approved for use in all Cummins® engines. Reference the engine specific Owners/Operations and Maintenance Manual for approved viscosity grades and specific recommendations for operating in cold ambient temperatures.

Note : For ISX15 and QSX15 dual overhead camshaft engines, 15w-40 viscosity grade is required.

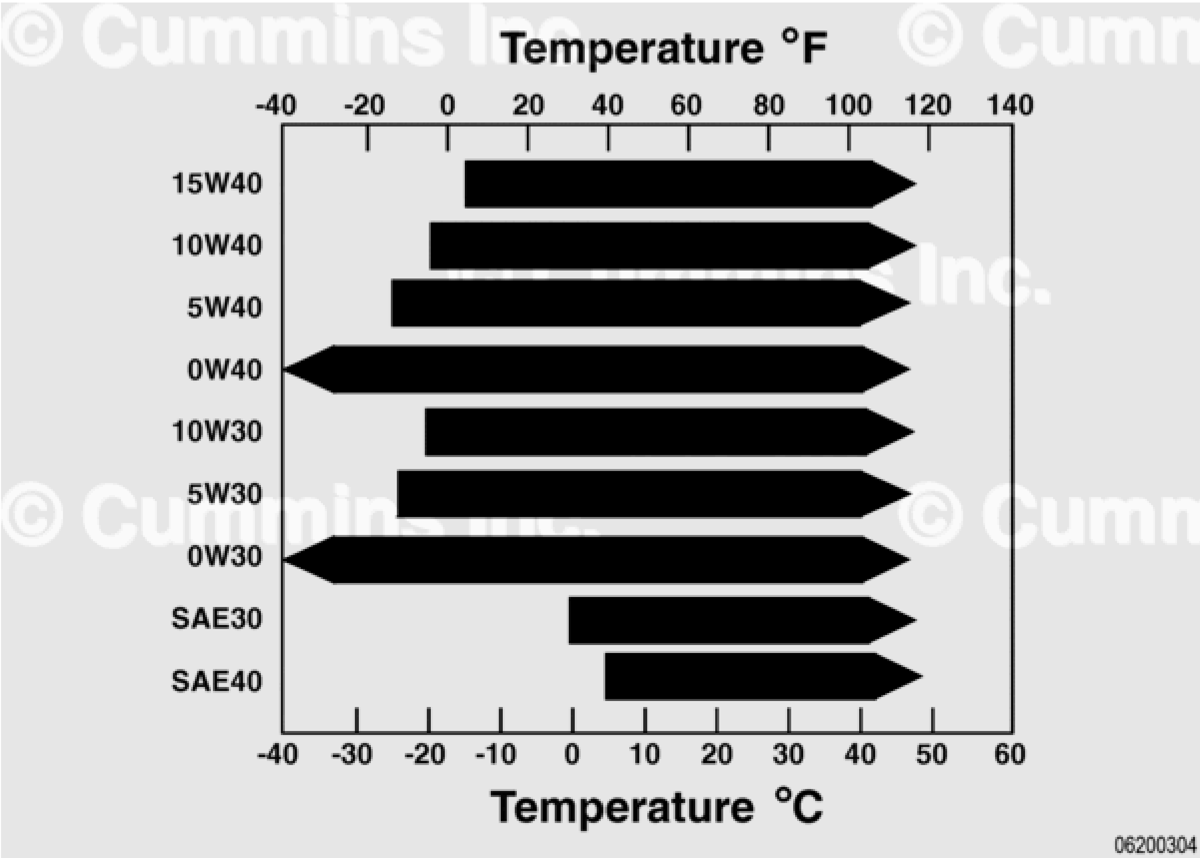

Figure 1: Recommended SAE Oil Viscosity Grades vs Ambient Temperatures